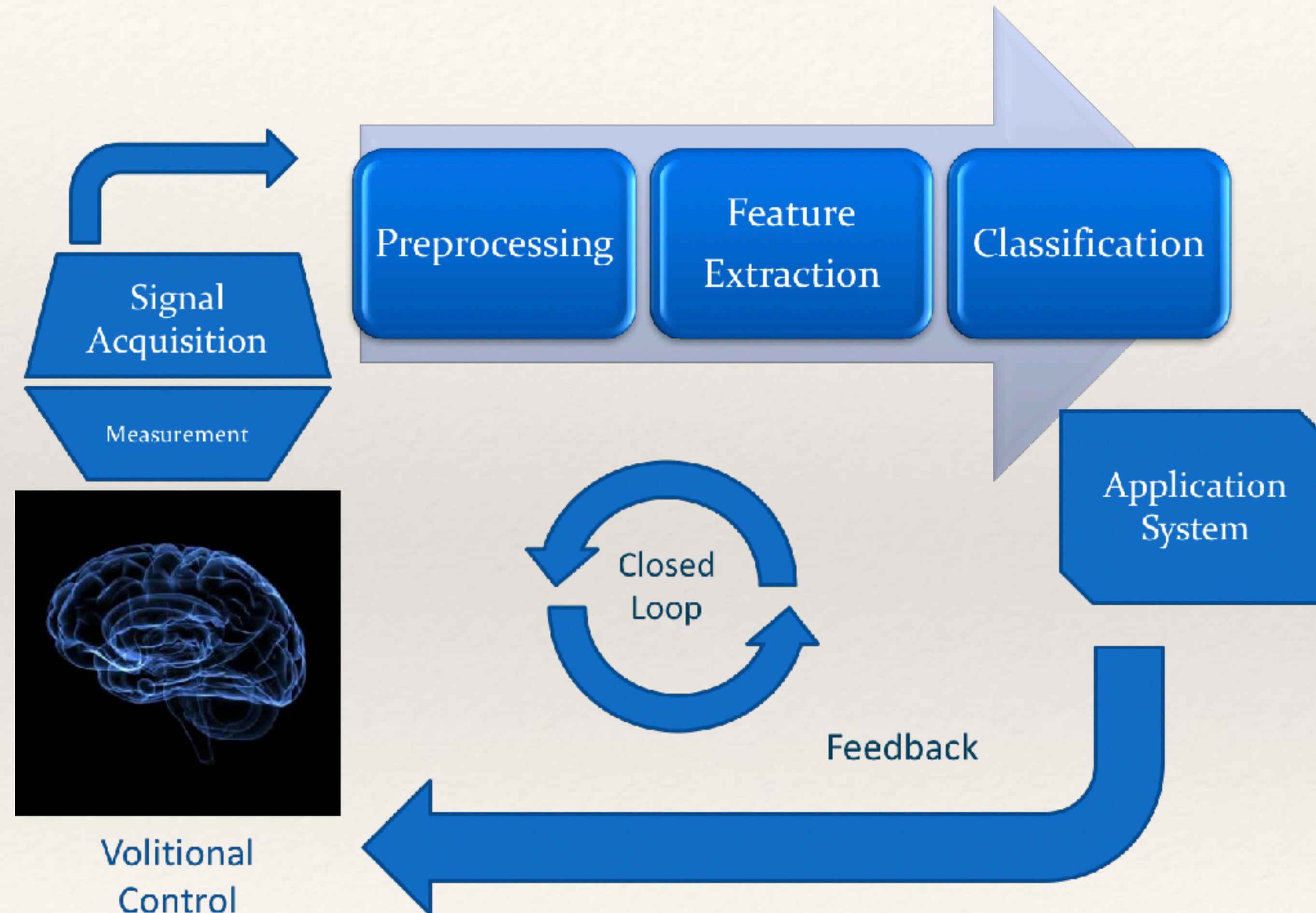


The materialization of Statistical Mathematics

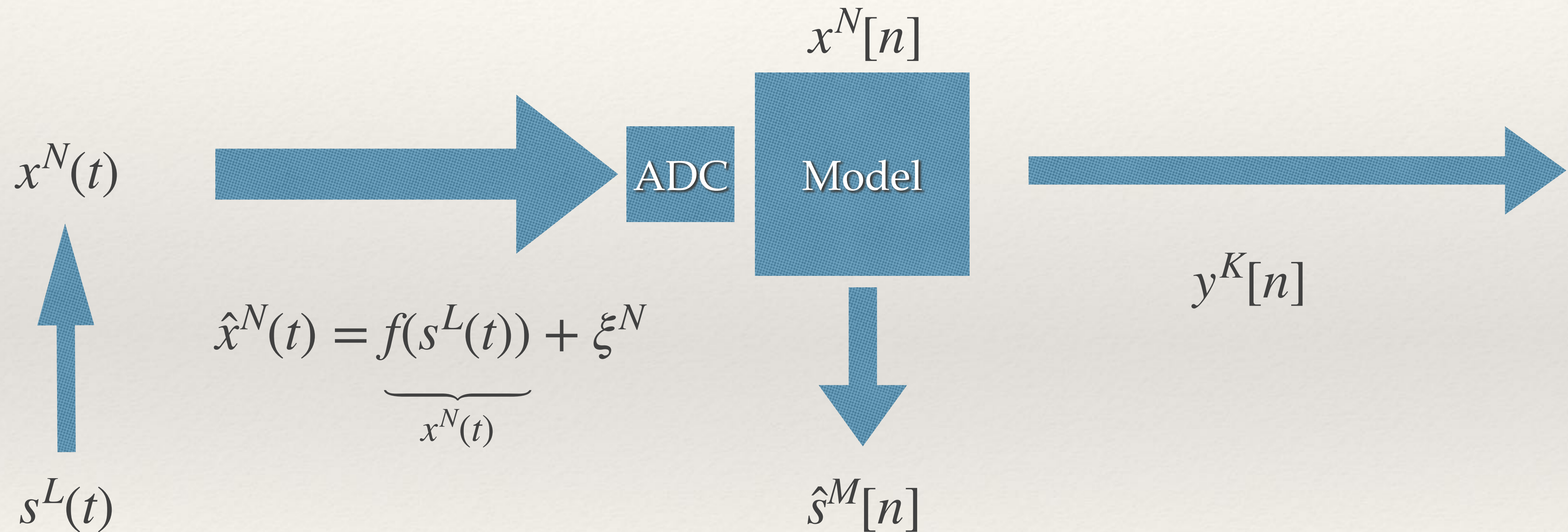
Machine Learning

“Learning” computers

Machine Learning Model



Engineering Inverse Problem



What is “learning”?

Model Free Approach
To determine the mapping
between some input X and
some output Y by
“adjusting” free parameters.

These free parameters are
adjusted based on an
Optimization Process



What is “learning”?

OPTIMIZATION PROBLEMS WITH CONSTRAINTS

- (1) A company wants to manufacture cylindrical aluminum cans with a volume of 1000cm^3 (1 liter). What should the radius and height of the can be to minimize the amount of aluminum used?

Volume of a Cylinder: $V = \pi r^2 h$

Surface Area of a Cylinder: $S = 2\pi r^2 + 2\pi r h$

We wish to optimize (minimize) the surface area $S = 2\pi r^2 + 2\pi r h$ with respect to the constraint $\pi r^2 h = 1000$. From this latter condition we have $h = \frac{1000}{\pi r^2}$, which gives

$$S = 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2} \right) = 2\pi r^2 + \frac{2000}{r}.$$

Differentiating,

$$S' = 4\pi r - \frac{2000}{r^2}.$$

Observe that $r = 0$ is a critical point. We solve the equation $S' = 0$, giving

$$S' = 0$$

$$4\pi r - \frac{2000}{r^2} = 0$$

$$4\pi r = \frac{2000}{r^2}$$

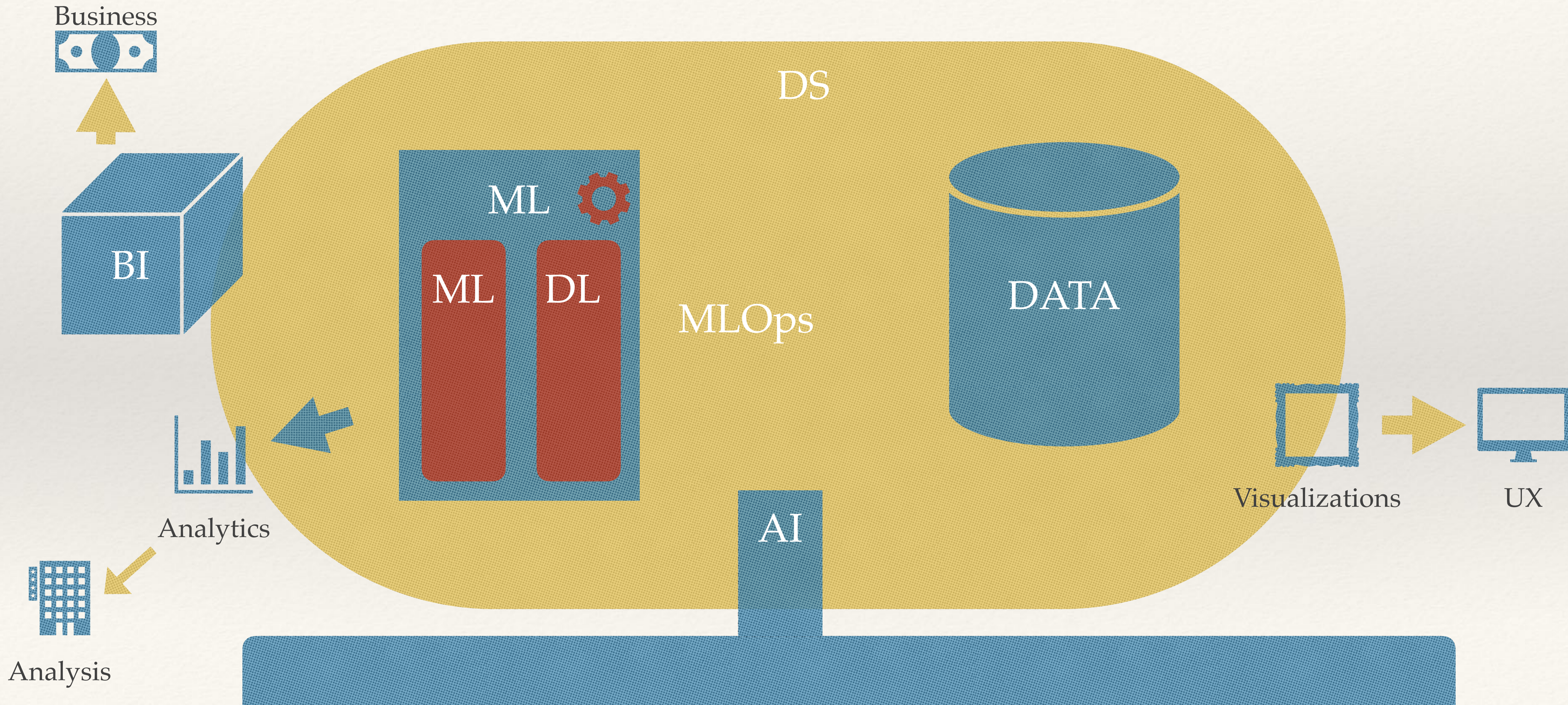
$$4\pi r^3 = 2000$$

$$r^3 = \frac{500}{\pi}$$

$$r = \left(\frac{500}{\pi} \right)^{1/3}.$$

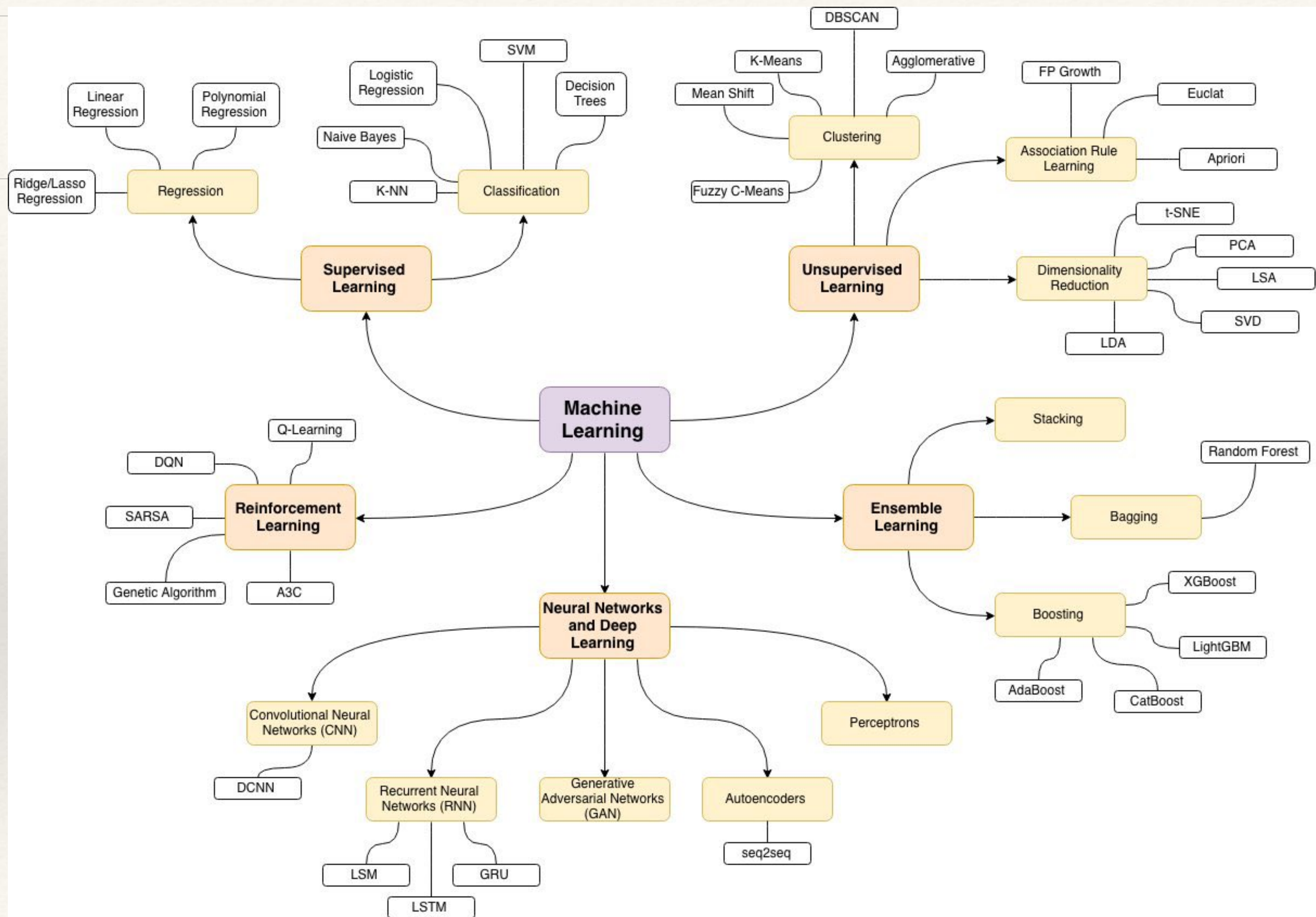
The derivative S' is negative for r in the interval $(0, (\frac{500}{\pi})^{1/3})$ and positive for $r > (\frac{500}{\pi})^{1/3}$, so we see that $r = (\frac{500}{\pi})^{1/3}$ is a minimum. For this value of r the equation $h = \frac{1000}{\pi r^2}$ gives $h = \frac{1000}{\pi (\frac{500}{\pi})^{2/3}}$.

Terminology

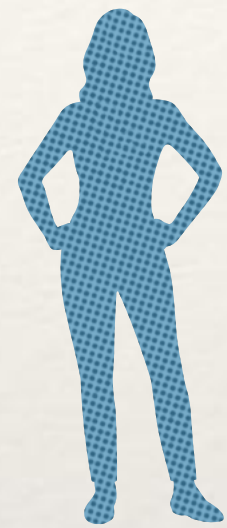


DL Map

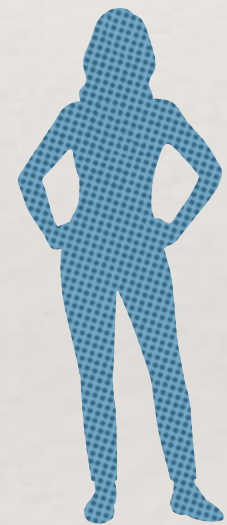




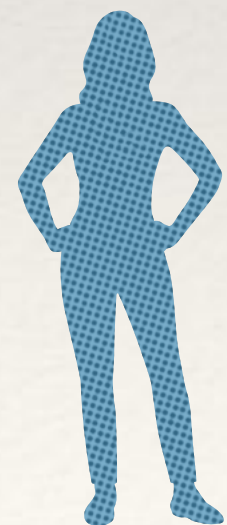
Roles



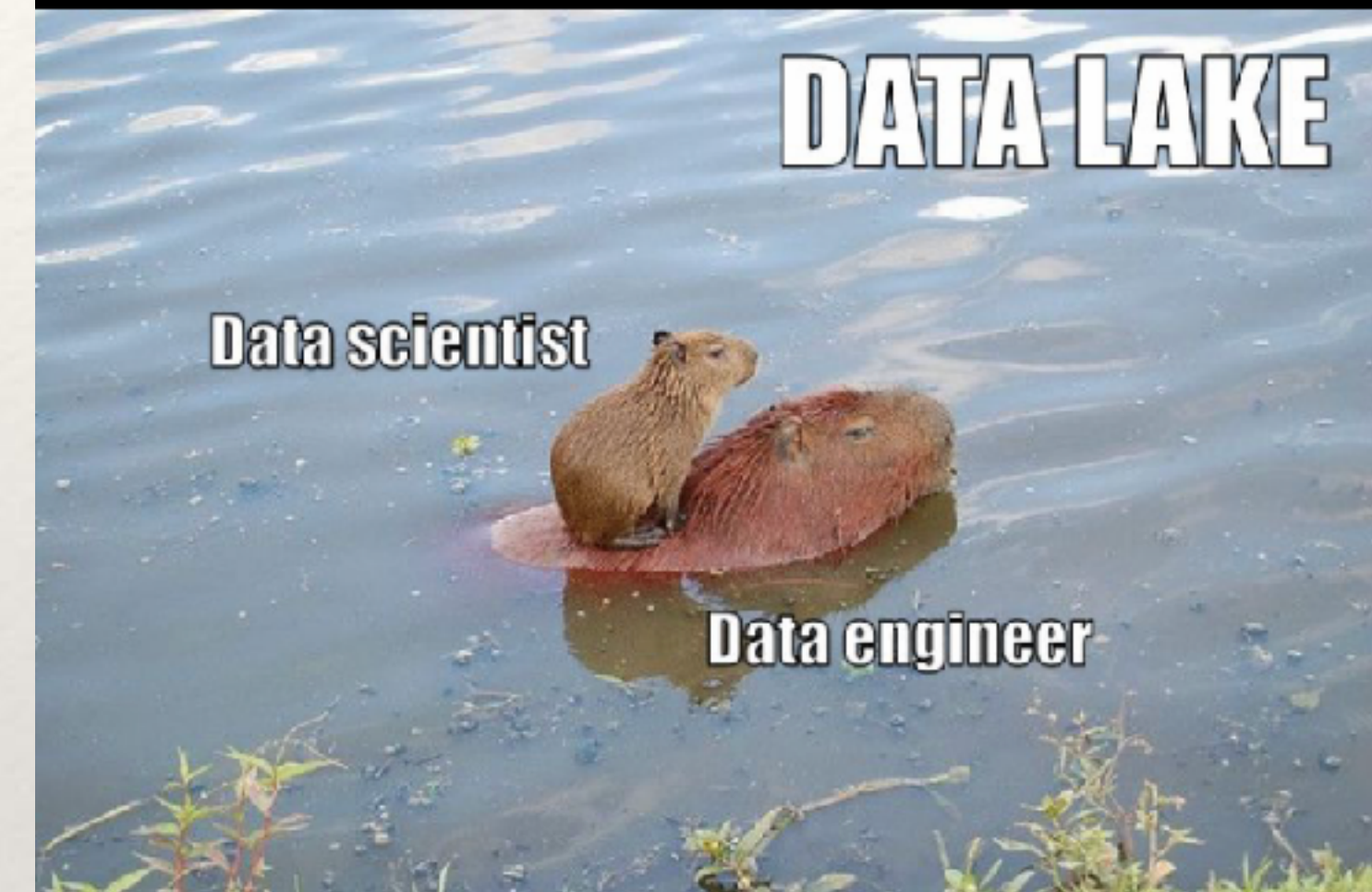
Data Engineer



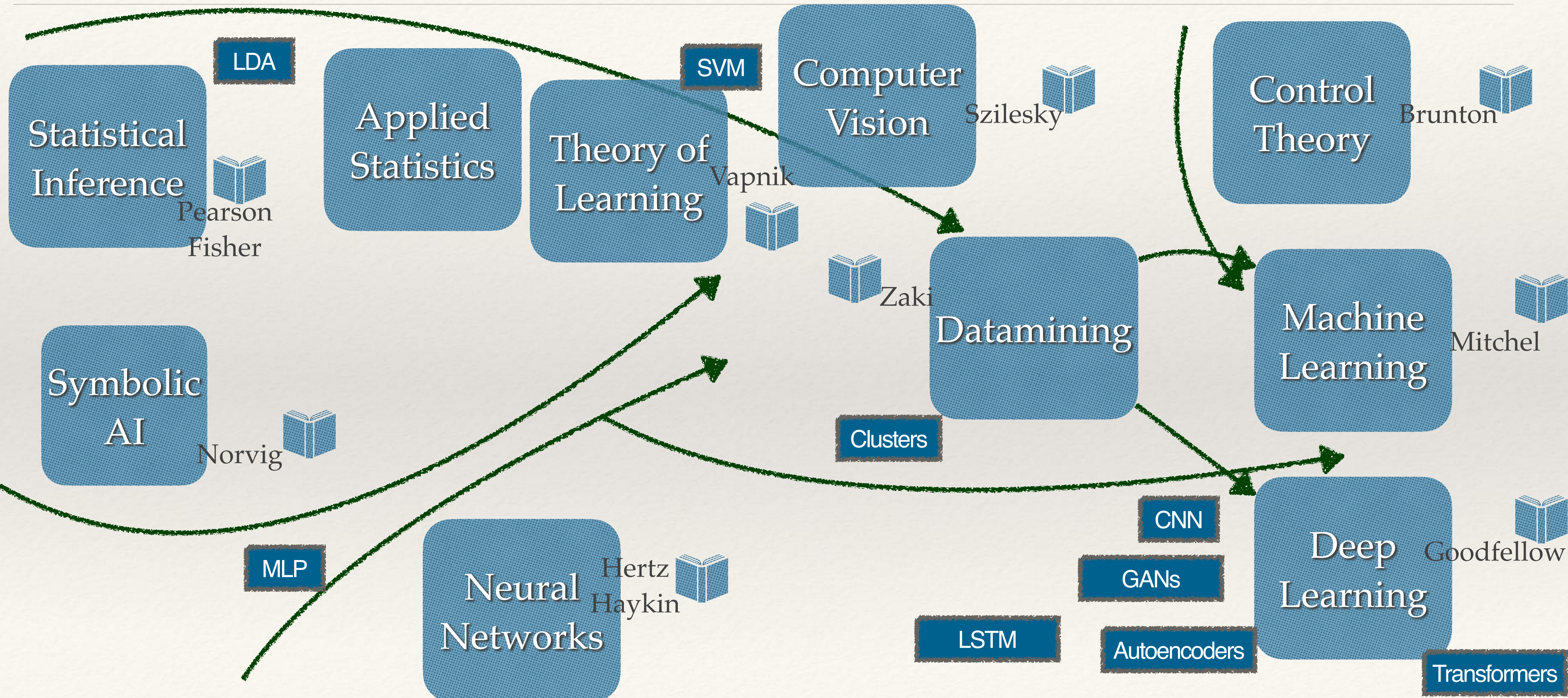
Data Scientist



Data Analyst



Evolving Artificial Intelligence



Emerging Technologies

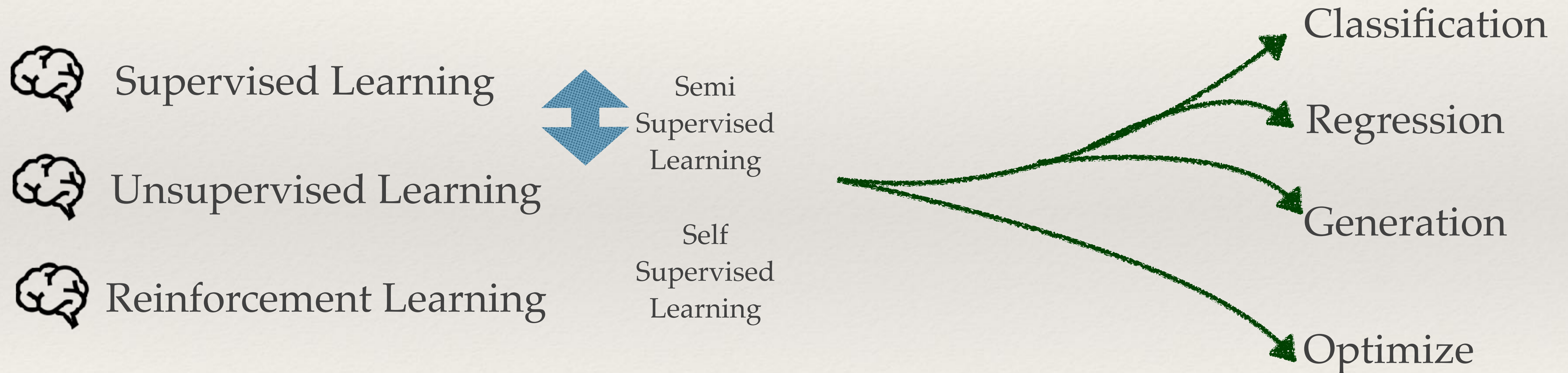
- ❖ GANs
- ❖ Explainable AI
- ❖ Edge AI
- ❖ AI PaaS
- ❖ Adaptive ML
- ❖ Emotion AI

Gartner Hype Cycle for Emerging Technologies, 2019



Learning by Data

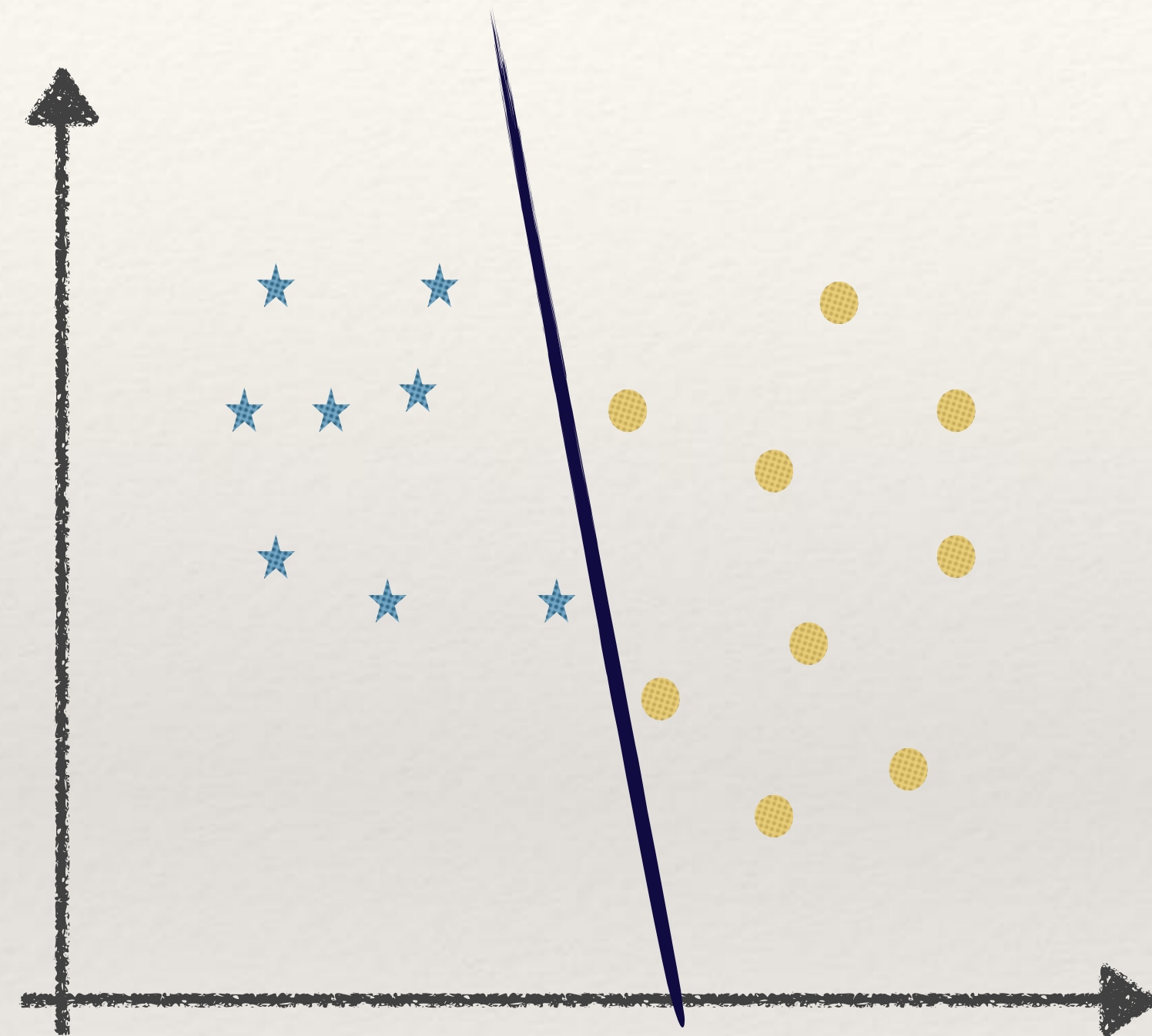
In ML there are three approaches to force a machine to “learn”



Machine Learning Techniques

SVM

LDA

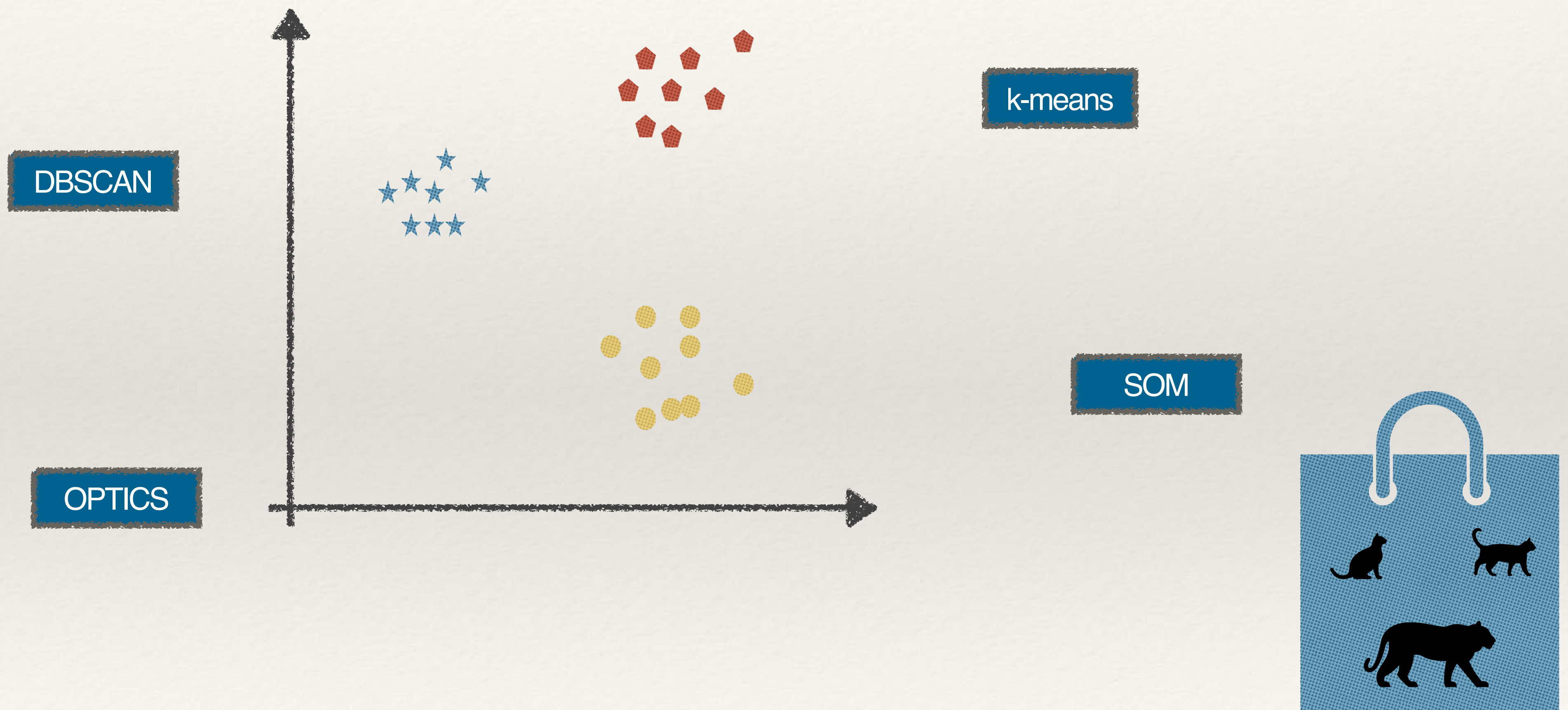


LogReg

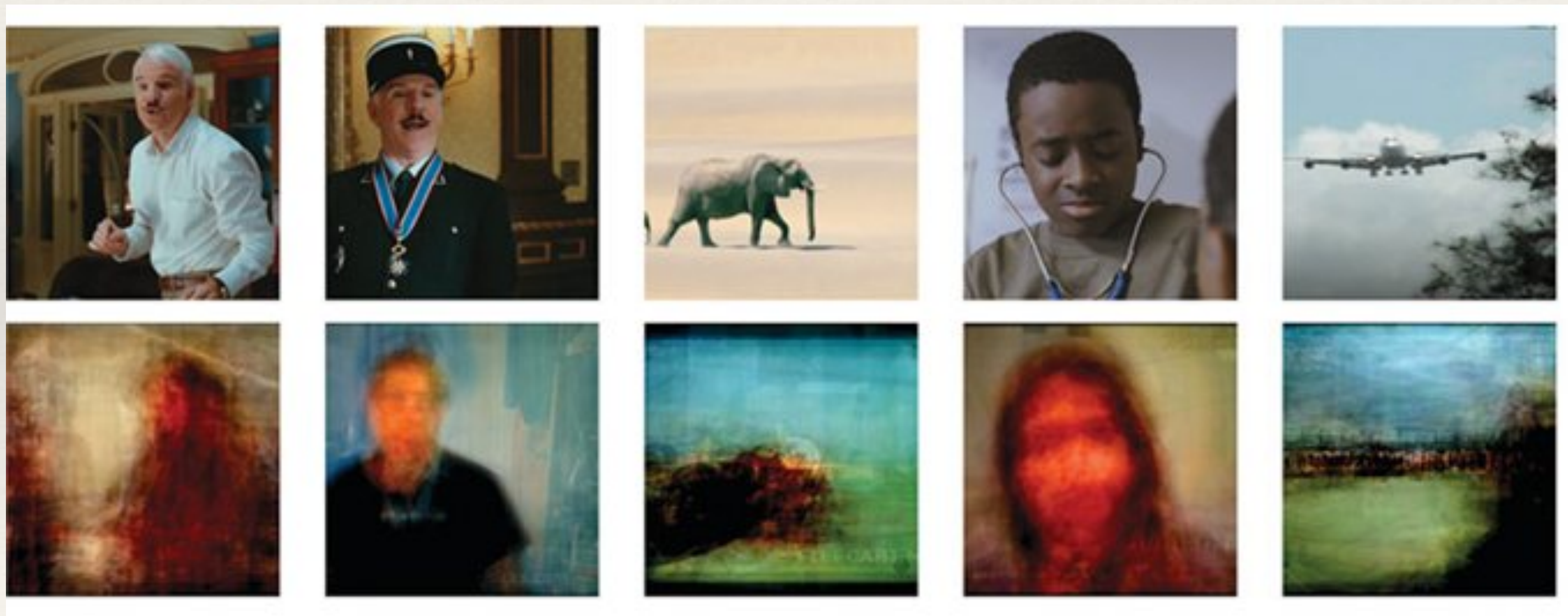
kNN

Autoregressive

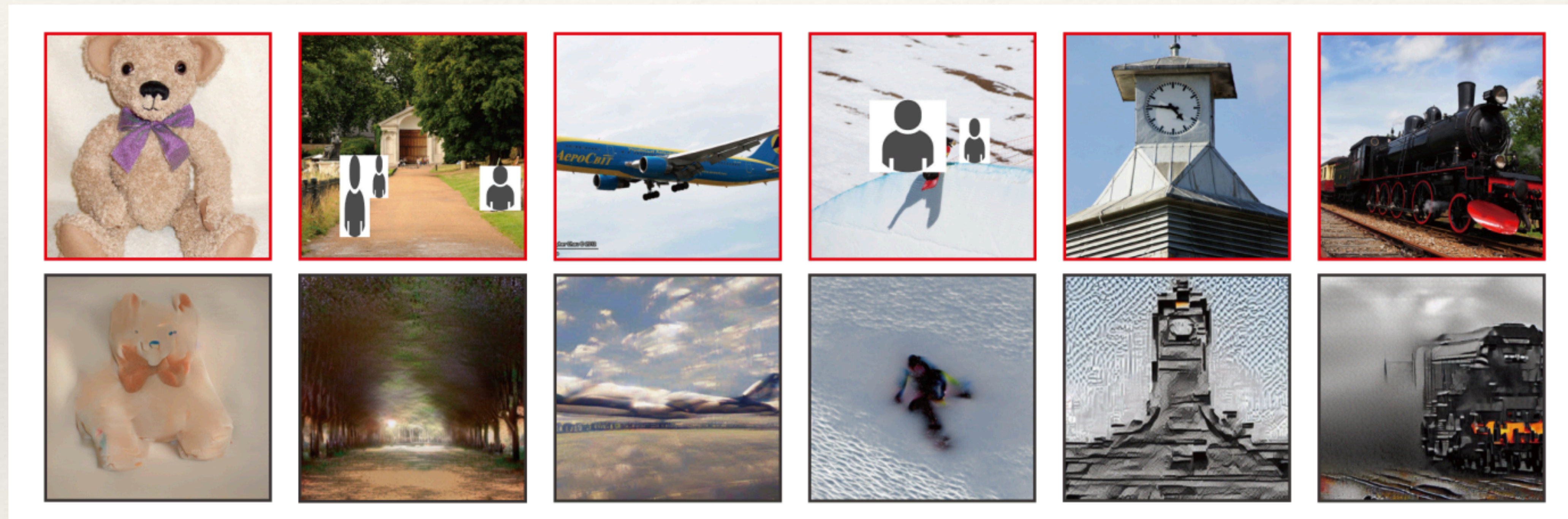
Machine Learning Techniques



Watching Dreams?



Watching Thoughts?



High-resolution image reconstruction with latent diffusion models from human brain activity, Takagi, Nishimoto 2022